

Chapter 4 Lab: Azure Storage Services Implementing Azure Storage

Task 1: Create a Storage Account

A **Storage Account** in Azure is the foundation for storing data objects in the cloud, providing a secure, scalable, and highly available environment to store files, blobs, queues, tables, and disks that can be accessed over the internet or within an Azure Virtual Network. In this task, I created a new resource group called **“RG-StorageLab”** using **Azure Subscription 1** and selected the region **South Africa North**. The resource group helps organize and manage all related storage resources efficiently. The chosen region is geographically close to my location, which reduces latency, ensures compliance with local regulations, and improves overall performance while optimizing costs. This configuration ensures reliable and efficient storage management within the Azure environment.

The screenshot shows the Azure portal interface for creating a new resource group. The browser address bar displays `https://portal.azure.com/#create/Microsoft.ResourceGroup`. The page title is "Create a resource group". Below the title, there are tabs for "Basics", "Tags", and "Review + create". The "Basics" tab is active. A description of a resource group is provided: "Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)". Three dropdown menus are visible: "Subscription" (selected: Azure subscription 1), "Resource group name" (text input: RG-StorageLab), and "Region" (selected: (Africa) South Africa North). At the bottom, there are three buttons: "Previous", "Next", and "Review + create" (highlighted in blue).

Home

Create a storage account ...

manage your storage account together with other resources.

Subscription *	Azure subscription 1
Resource group *	RG-StorageLab
	Create new
Instance details	
Storage account name * ⓘ	storagentibo123
Region * ⓘ	(Africa) South Africa North
	Deploy to an Azure Extended Zone
Preferred storage type	Azure Blob Storage or Azure Data Lake Storage Gen 2
	i This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more
Primary workload ⓘ	Select...
	Pick the closest match to your workload to get a recommended configuration built on best practices. You can edit this configuration anytime. See comparison table
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account)
	☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally-redundant storage (LRS)

The reason I choice Azure Blob Storage or Azure Data Lake Storage Gen 2 is because it is cost effective, general-purpose unstructured data with a flat namespace, enhanced security, and analytics optimization.

Task 2: Create Blob Storage Container

After deployment, I accessed the Storage Account to manage its resources. I navigated to the **Containers** section, where I was able to create a new container. The container was created using the instructed name, and the access level was left as private by default to ensure that no anonymous access is allowed. This configuration enhances security by restricting access to authorized users only, ensuring that the data stored within the container remains protected.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

koportalntbo@gmail.com
DEFAULT DIRECTORY (KOPORALNTBO)

Home > storagentibo123_1775158676403 | Overview > storagentibo123

storagentibo123 | Containers ☆ ...

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

File shares

Queues

Tables

Security + networking

Data management

Continue

Add or remove favorites by pressing CTRL+SHIFT+F

Search containers by prefix

Only show active containers

Showing all 1 items

☐	Name	Last modified	Anonymous access level	Lease state	
☐	 \$logs	4/2/2026, 9:38:45 PM	Private	Available	...

Home > storagentibo123_1775158676403 | Overview > storagentibo123

storagentibo123 | Containers ☆ ...

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

New container

Name *

studentfiles

Home > storagentibo123_1775158676403 | Overview > storagentibo123

storagentibo123 | Containers ☆ ...

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

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Tables

Security + networking

Data management

Continue

Search containers by prefix

Only show active containers

Showing all 2 items (2 selected)

☒	Name	Last modified	Anonymous access level	Lease state	
☒	 \$logs	4/2/2026, 9:38:45 PM	Private	Available	...
☒	 studentfiles	4/2/2026, 9:41:55 PM	Private	Available	...

TASK 3: UPLOAD FILE TO BLOB STORAGE

Upload any file inside the newly created container (Blob storage). When uploaded it will reveal details about the file including name, size, last modified date.

Copilot



koporalntibo@gmail.com
DEFAULT DIRECTORY (KOPORAL...



columns

Upload blob





Drag and drop files here
or
[Browse for files](#)

Select an existing container

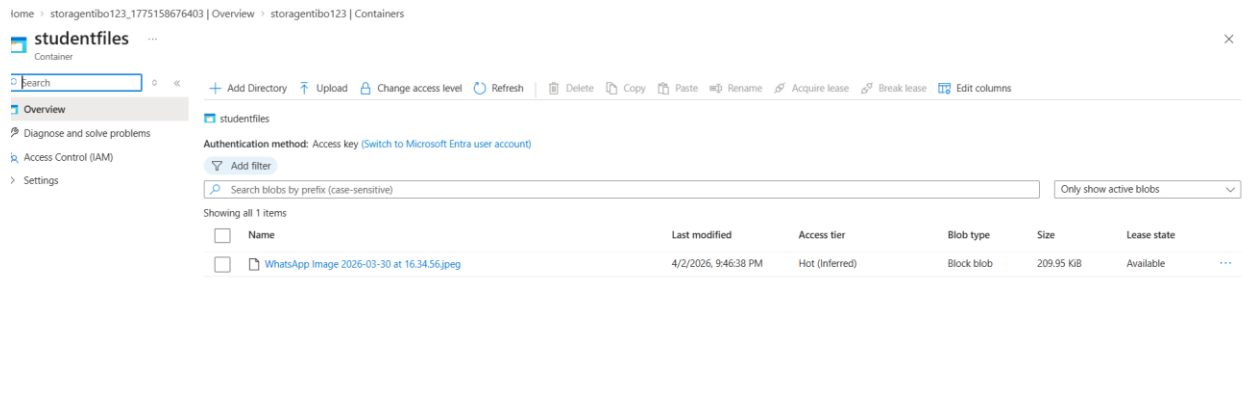
[Create new](#)

☐ Overwrite if files already exist

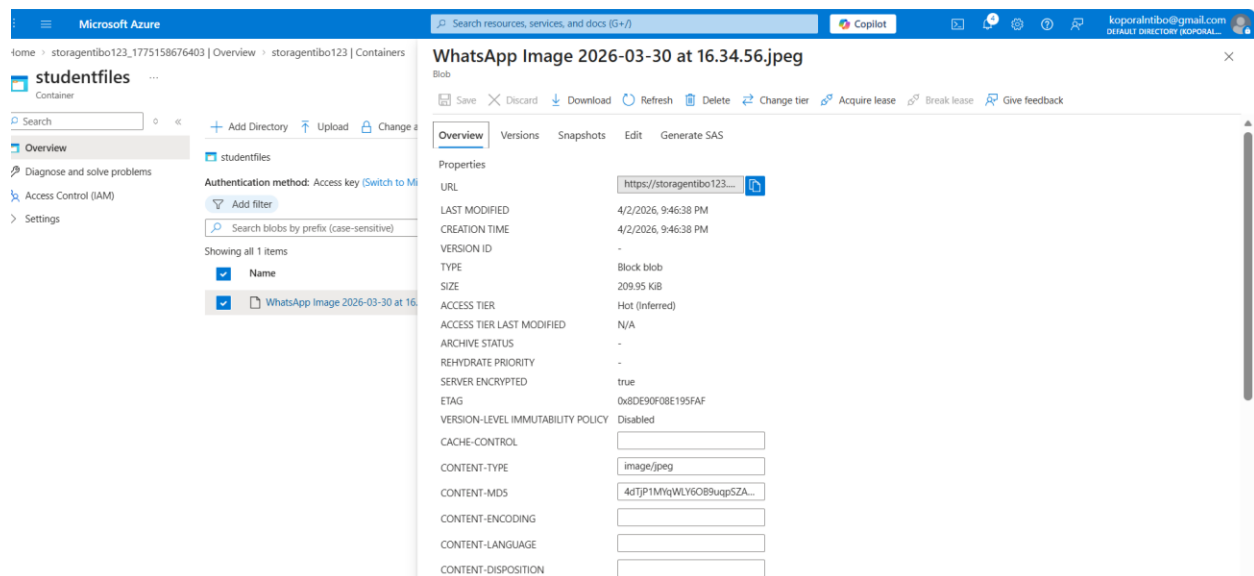
Advanced

Upload

 [Give feedback](#)



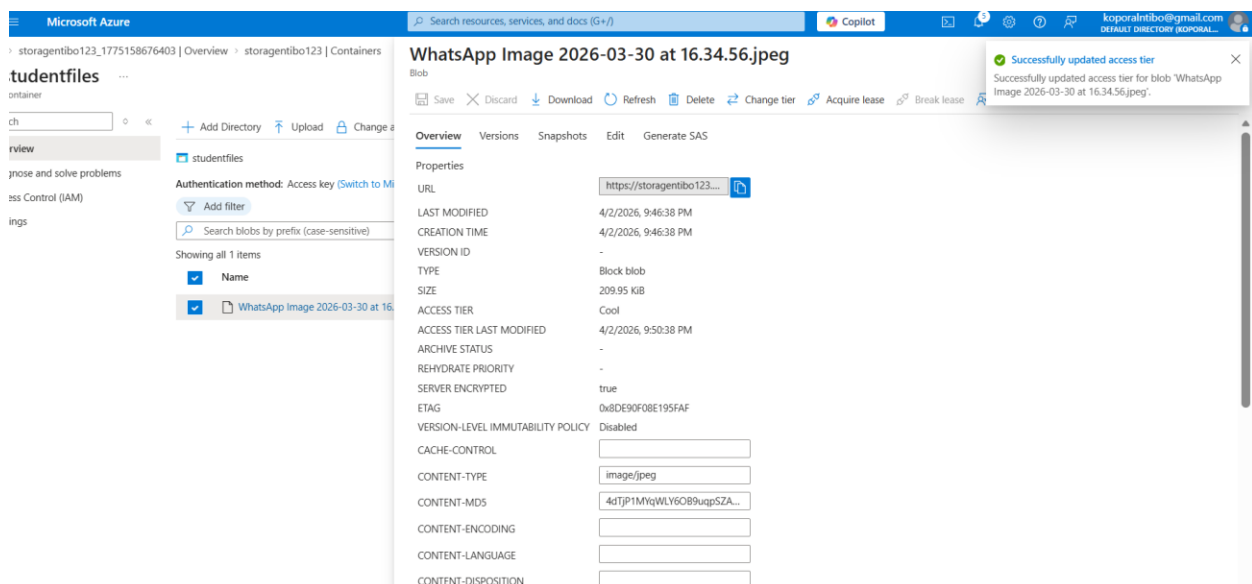
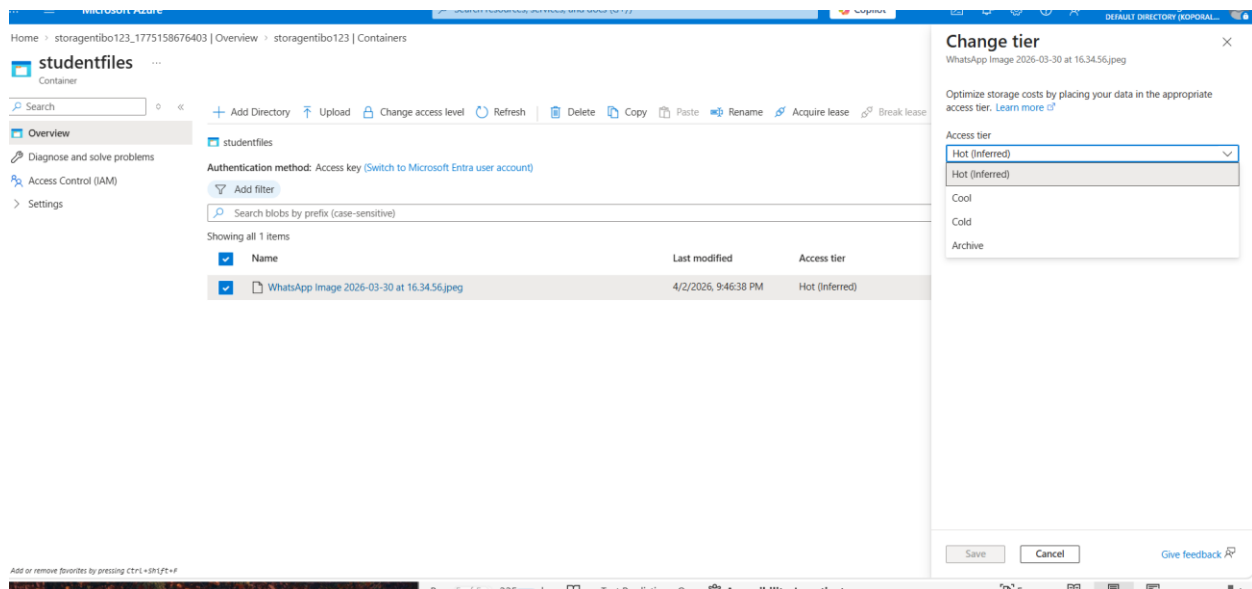
Details about uploaded file.



Task 4: Change Storage Access Tier

- Hot - frequently accessed data
 - Cool - occasionally accessed
 - Archive - rarely accessed
- Changing Azure storage access tiers optimizes costs based on data usage patterns.

Changing from Hot access tier to Cool access tier in Azure reduces storage costs for data that is infrequently accessed.



Home > storagentibo123_1775158676403 | Overview > storagentibo123 | Containers

studentfiles ...

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

studentfiles

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☒	Name	Last modified	Access tier	Blob type	Size	Lease state	
☒	 WhatsApp Image 2025-03-30 at 16.34.56.jpeg	4/2/2025, 9:46:38 PM	Cool	Block blob	209.95 KiB	Available	...

Task 5: Attach Managed Disk To Virtual Machine

Create a virtual machine

[Help me create a VM optimized for high availability](#)[Help me choose the right VM size for](#)

i There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host

☐

i Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

128 GiB (E10)

i Some images are, by default, smaller than the selected OS disk size. [Click here to learn how to expand your disk partition size after you create your VM.](#)

OS disk type *

Standard SSD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM

☒

[< Previous](#)

[Next : Networking >](#)

[Review + create](#)

The disk size we are instructed to use is 32 GB, but my account has sizes starting from 128 Gb upwards. Standard SSD = best for web servers, lightly used enterprise applications and development and testing.

Locally-redundant storage (data is replicated within a single datacenter)

Premium SSD

Best for production and performance sensitive workloads

Standard SSD

Best for web servers, lightly used enterprise applications and dev/test

Standard HDD

Best for backup, non-critical, and infrequent access

Zone-redundant storage (data is replicated to three zones)

Premium SSD

Best for the production workloads that need storage resiliency against zone failures

Standard SSD

Best for web servers, lightly used enterprise applications and dev/test that need storage resiliency against zone failures

Task 6: Create Azure File Share

Home > storagegwama123 | File shares

New file share ...

Basics Backup Review + create

Name *

sharedfiles

Access tier *

Transaction optimized

Performance

Maximum IO/s ⓘ

20000

Maximum capacity

100 TiB

ⓘ To use the SMB protocol with this share, check if you can communicate over port 445. These scripts for [Windows clients](#) and [Linux clients](#) can help. Learn how to [circumvent port 445 issues](#).

Deleting the resource group “RG-StorageLab” will remove all resources, creating space and lower costs in the cloud.